INCH-POUND
MIL-DTL-16377/17B(SH)
30 June 2014
SUPERSEDING
MIL-F-16377/17A(SH)
24 December 1981

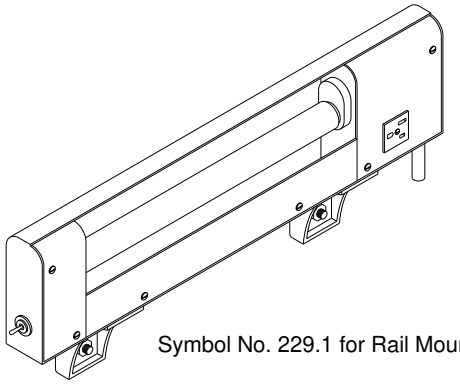
DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT AND SSL EQUIVALENT, DETAIL LIGHTING FOR BERTHS, 8 WATTS, 120 VOLTS, 60 HERTZ, SYMBOLS 229.1, 231.1, 232.1, 229.1SSL, 231.1SSL, AND 232.1SSL

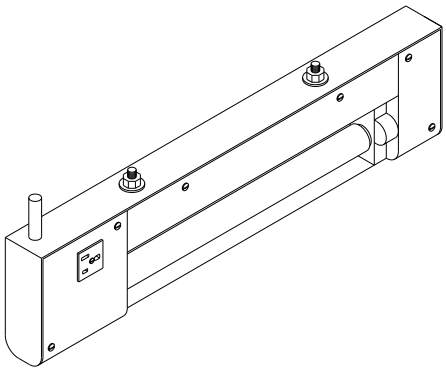
This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

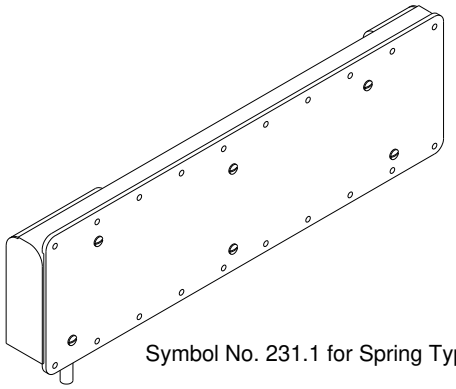
All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.



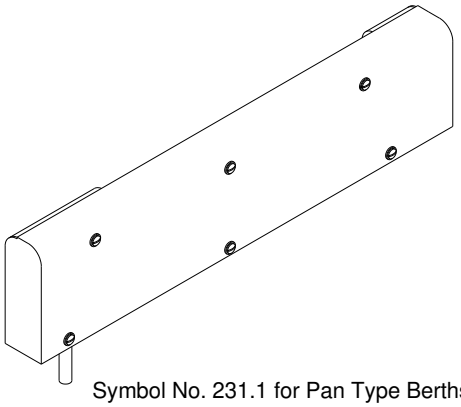
Symbol No. 229.1 for Rail Mounting



Symbol No. 232.1 for Bulkhead or Bar Mounting



Symbol No. 231.1 for Spring Type Berths



Symbol No. 231.1 for Pan Type Berths

FIGURE 1. Typical view of symbols 229.1, 231.1, and 232.1.

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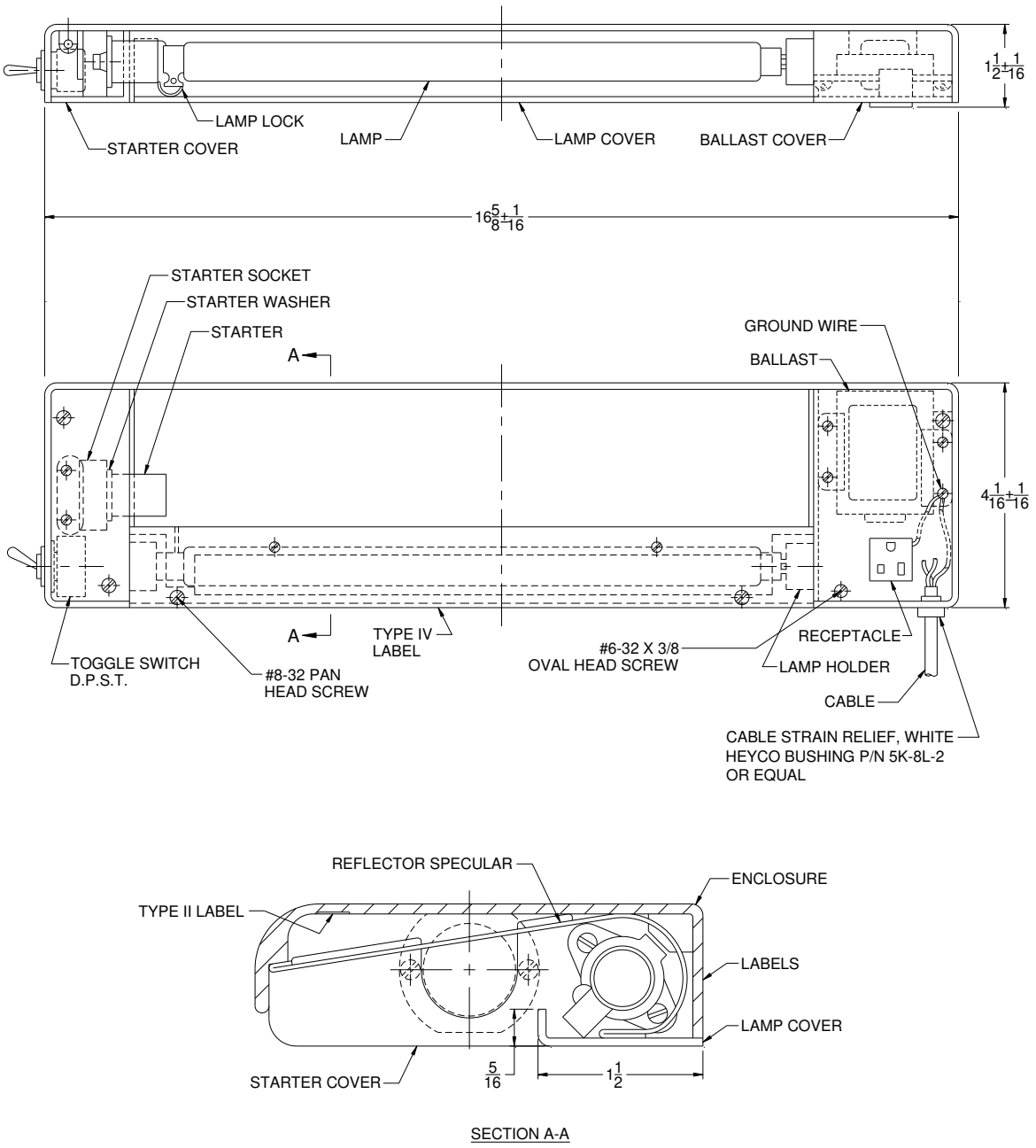


FIGURE 2. Typical plan and section views of basic light for symbols 229.1, 231.1, and 232.1.

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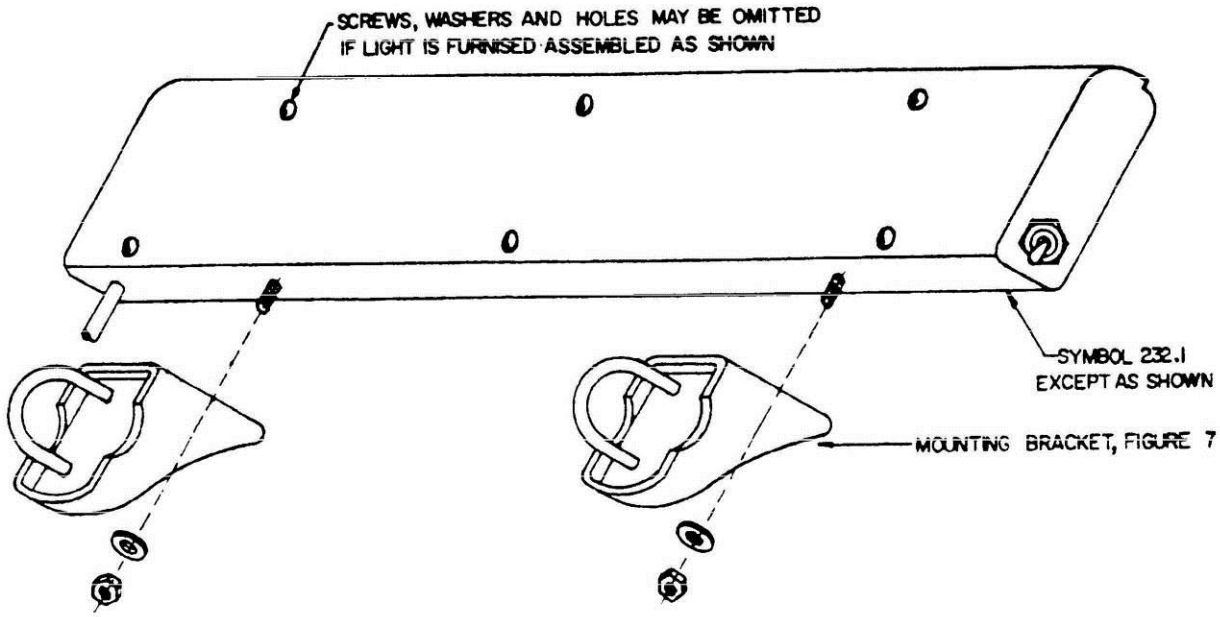


FIGURE 3. Typical view of symbol 229.1.

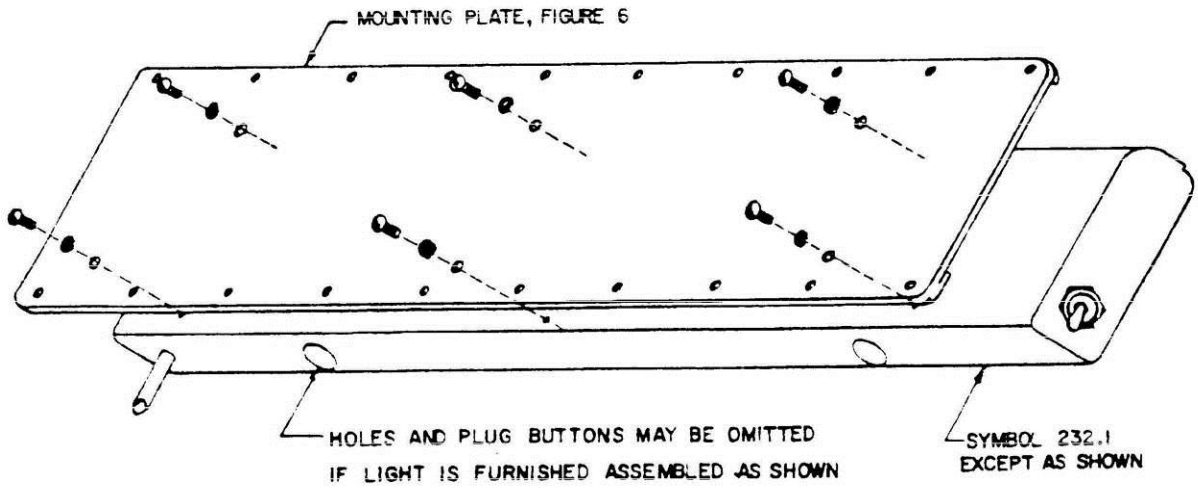


FIGURE 4. Typical view of symbol 231.1.

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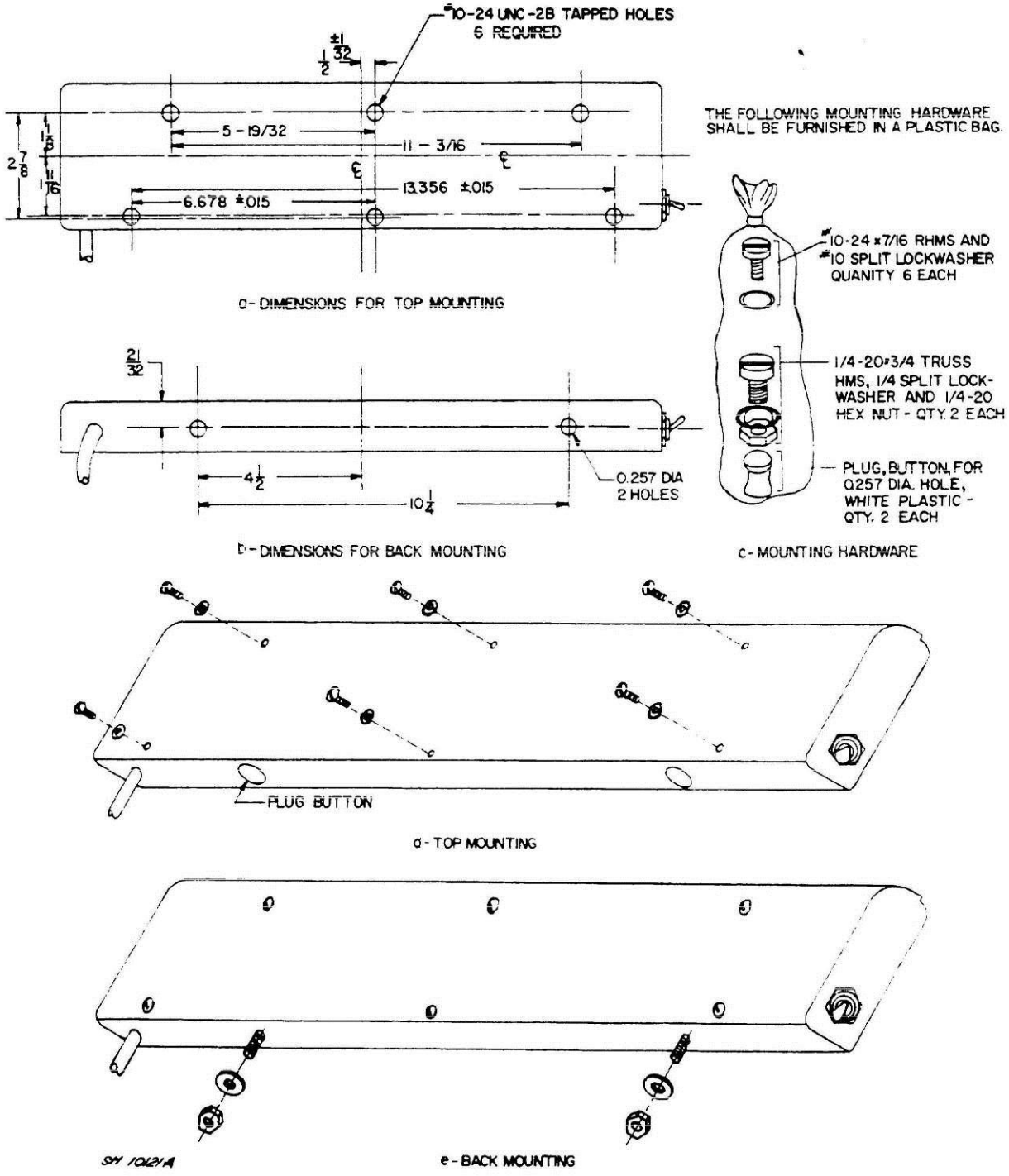


FIGURE 5. Typical view of symbol 232.1.

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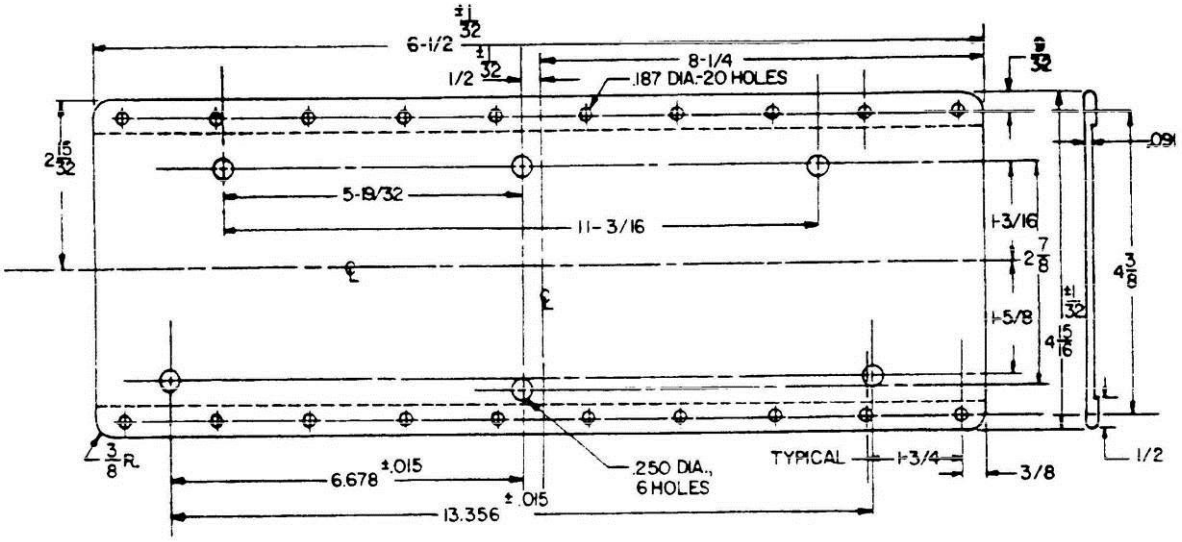


FIGURE 6. Mounting plate for symbol 231.1.

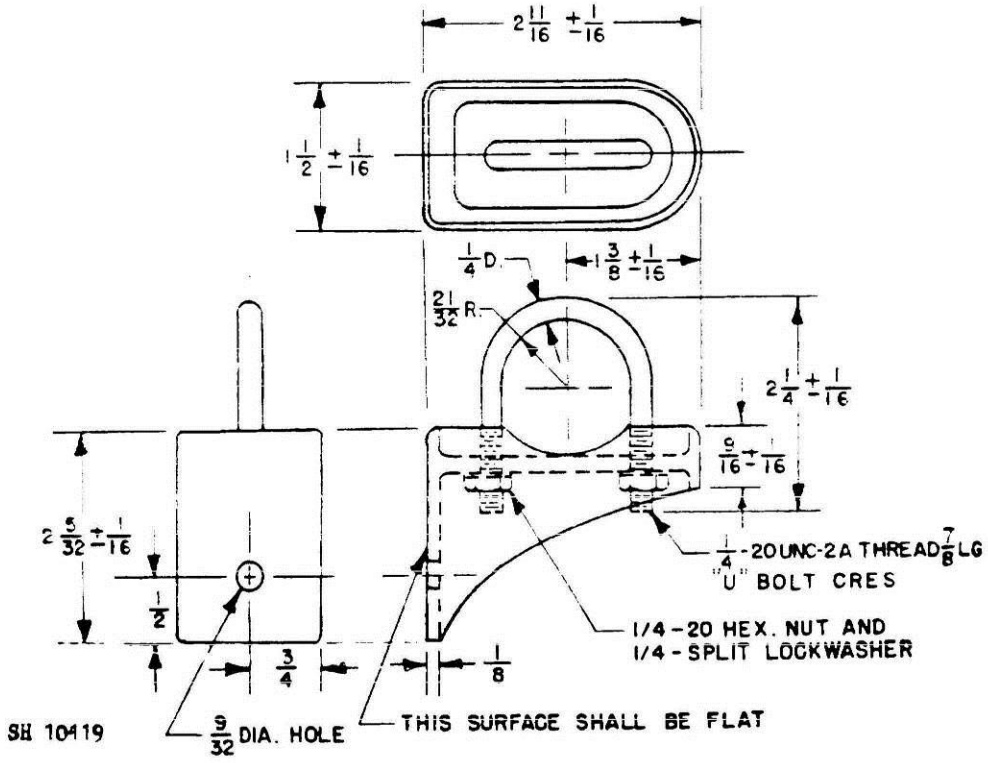


FIGURE 7. Mounting bracket for symbol 229.1.

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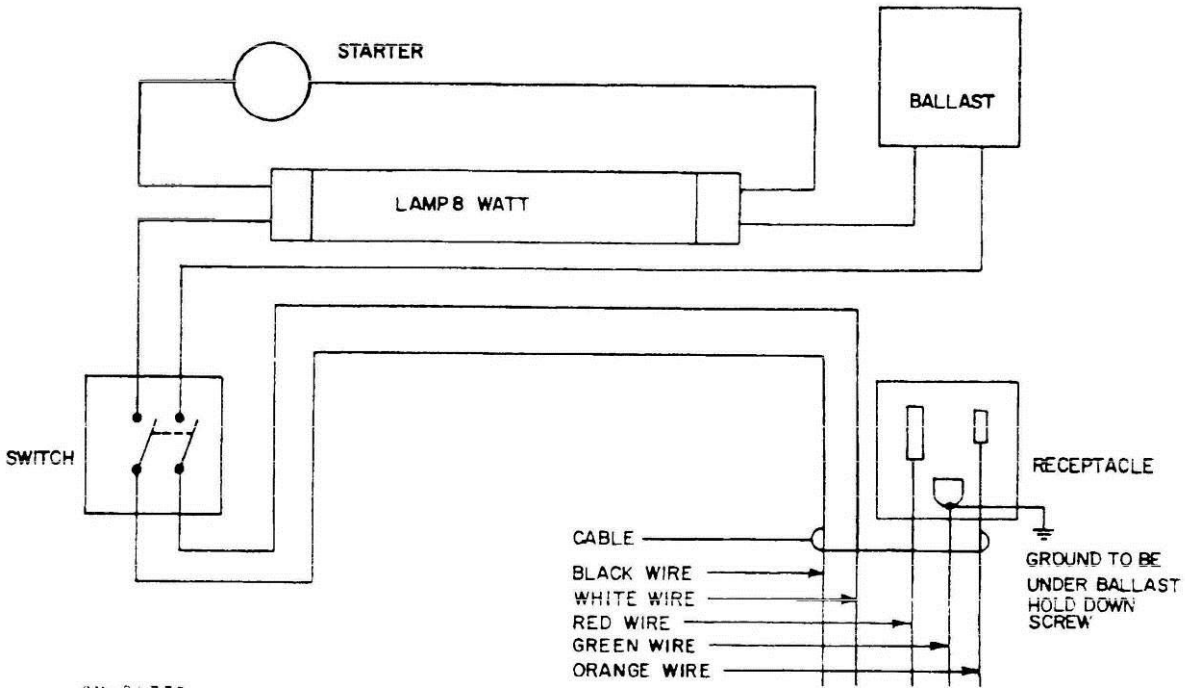
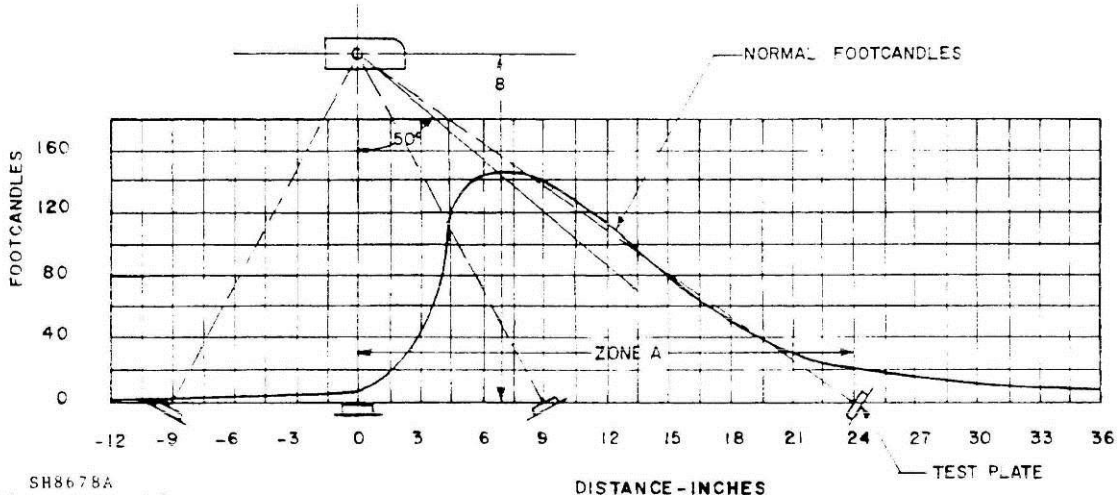


FIGURE 8. Wiring diagram.



Footcandle values within Zone "A" shall be not less than 90 percent of the value shown hereon. These results shall be met after correction to 380 lumens output for both lamp and ballast has been made.

Voltage:	115 volts, 60 hertz.
Ambient temperature:	76 °F (24.4 °C).
Lamps:	Lamp furnished with light shall be used during test.
Ballast:	Ballast furnished with light shall be used during test.
Lamp correction:	Distribution curves based on 380 lumens per lamp.

FIGURE 9. Footcandle distribution curves.

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FLUORESCENT REQUIREMENTS:

CLASSIFICATION:

Type: 1
Class: 1

SYMBOL NUMBERS:

229.1: Complete lighting fixture assembly, for rail mounting, with one 8-watt cool white fluorescent lamp.

231.1: Complete lighting fixture assembly, for spring type berths (bunks), with one 8-watt cool white fluorescent lamp.

232.1: Complete lighting fixture assembly, for pan type berths, bulkhead or bar mounting, with one 8-watt cool white fluorescent lamp.

DESIGN:

Electrical rating:	8 watts maximum, 120 volts, 60 hertz.
Dimensions and configurations:	See figures 1 through 7.
Photometric data:	See figure 9.
Lampholder spacing:	See figure 21 of MIL-DTL-16377.
Wiring diagram:	See figure 8.
Material:	Material for enclosure, lamp cover, mounting plate and mounting brackets shall be aluminum or plastic Type MAI-60 in accordance with ASTM D5948.
Finish:	Finish shall comply with 3.8.3.4 of MIL-DTL-16377, except that color shall be beige conforming to Color Number 26586 of FED-STD-595.
Cable strain relief:	Strain relief for cable shall be provided as shown on figure 2.
Grounding:	Aluminum enclosure: See figure 2; Plastic enclosure: The reflector, ballast frame and exterior metal parts of switch shall be grounded to receptacle ground contact.
Receptacle:	Receptacle shall be Leviton part number 1374-051 (ivory), or equal except that the ground contact shall be nickel-plated 0.0003 inch thick.
Cable:	10 feet of Type LSMHOF-7 in accordance with MIL-DTL-24643/7.
Weight:	4 pounds 2 ounces maximum.
Reflector:	Reflector shall be made of specular, pre-finished lighting sheet.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figure 2.

a. One Type I.

b. One Type IV. The information to be shown on the Type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by a list of all standard parts shown in table I by name and stock number only.

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SSL REQUIREMENTS:

CLASSIFICATION:

Type: III
Class: 2

SYMBOL NUMBER:

229.1SSL: Complete lighting fixture assembly, for rail mounting, with LED luminaire and white diffusing window assembly.
231.1SSL: Complete lighting fixture assembly, for spring type berths (bunks), with LED luminaire and white diffusing window assembly.
232.1SSL: Complete lighting fixture assembly, for pan type berths, bulkhead or bar mounting, with LED luminaire and white diffusing window assembly.

DESIGN:

Electrical rating:	8 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 1 through 7.
Photometric data:	Initial light output shall be within +20 percent and -10 percent of figure 9.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.17 of MIL-DTL-16377.
Material:	Material for enclosure, lamp cover, mounting plate and mounting brackets shall be aluminum or plastic Type MAI-60 in accordance with ASTM D5948.
Finish:	Finish shall comply with 3.8.3.4 of MIL-DTL-16377, except that color shall be beige conforming to Color Number 26586 of FED-STD-595.
Cable strain relief:	Strain relief for cable shall be provided as shown on figure 2.
Grounding:	Aluminum enclosure: See figure 2; Plastic enclosure: The reflector, ballast frame and exterior metal parts of switch shall be grounded to receptacle ground contact.
Receptacle:	Receptacle shall be Leviton part number 1374-051 (ivory), or equal except that the ground contact shall be nickel-plated 0.0003 inch thick.
Wiring:	Shall maintain same control scheme for light output as legacy fluorescent fixture. Refer to figure 8.
Cable:	10 feet of Type LSMHOF-7 in accordance with MIL-DTL-24643/7.
Weight:	4 pounds 2 ounces maximum.

Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 3:

a. Two Type I.
b. One Type IV. The information to be shown on the Type IV label shall be “USE THE FOLLOWING REPAIR PARTS ONLY:” followed by:
   (1) A list of all standard parts shown in table I by name and stock number only, if appropriate, or
   (2) A list of available replacement parts and stock numbers, or if unavailable, manufacturer’s part numbers.

Standard parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

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EXAMINATION AND TESTS:

Examinations and tests shall be as specified in MIL-DTL-16377 and the following:

First article inspection:
Complete fixtures: See table II for Type I and table III for Type III.
Conformance:
Comparison inspection: Same as first article inspection.
Conformance inspection: See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of military part numbers and general information is shown in table IV.

TABLE I. Standard Parts.

Name	Specification or Military Part Number	National Stock Number (NSN)	Remarks
Lamp	ANSI/ANSLG C78.81	6240-00-299-5546	Industry Number F8T5/CW
Lamp lock	M16377/40-001	6250-00-399-7039	
Lampholder	UL 496	6250-00-377-8590	SUPP. SA, FIG. SA2.1
Startersocket	UL 496	6250-00-295-2738	SUPP. SA, FIG. SA2.4
Starter	NEMA ANSI C78.180	6250-00-299-5962	Commercial Designation: FS5
Ballast	M16377/45-001	6250-00-295-2735	
Switch, toggle, DPST	MS16569-1	5930-00-719-9659	

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TABLE II. First article and comparison inspection for complete type I fixtures, symbols 229.1, 231.1, 232.1. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Shock	3.5.12	4.8.7	[2/]
Vibration	3.5.13	4.8.8	[2/]
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Noise	3.5.14	4.8.9	[2/]
Grounding circuit	3.6.13.3	4.8.11	
Light output	3.7.2	4.8.6.2	Footcandle
Magnetic permeability	3.4.8	4.8.18	

NOTES:

[1/] Requirement and test method paragraph numbers refer to MIL-DTL-16377.

[2/] In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE III. First article and comparison inspection for complete type III fixtures, symbols 232.1SSL, 229.1SSL, 231.1SSL. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Shock	3.5.12	4.8.7	[2/]
Vibration	3.5.13	4.8.8	[2/]
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Noise	3.5.14	4.8.9	[2/]
Continuity of grounding	3.6.13.4	4.8.12	
Photometrics	3.7.1	4.8.6	
Brightness	3.7.5	4.8.6.4	
Chromaticity	3.7.4	4.8.6.5	
Correlated color temperature (CCT), white light only	3.7.4.1.1	4.8.6.6	
Lumen maintenance	3.7.11	4.8.20	
Fail-safe circuit design	3.17	4.8.24	
Magnetic permeability	3.4.8	4.8.18	

NOTES:

[1/] Requirement and test method paragraph numbers refer to MIL-DTL-16377.

[2/] In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE IV. Part number and general information.

Military Part Number M16377/17-	Item Name	National Stock Number (NSN)	Remarks
229.1	Complete light, Symbol 229.1	6230-01-240-1105	1/ 2/ figure 3
231.1	Complete light, Symbol 231.1	6230-01-244-2936	1/ 2/ figure 4
232.1	Complete light, Symbol 232.1	6230-00-578-7548	1/ 2/ figure 5
232.1SSL	Complete fixture, solid state, Symbol 232.1SSL		
229.1SSL	Complete fixture, solid state, Symbol 229.1SSL		
231.1SSL	Complete fixture, solid state, Symbol 231.1SSL		
001	Mounting bracket for Symbol 229.1 - two brackets		1/ 2/ figure 7
002	Mounting plate for Symbol 231.1		1/ 2/ figure 6

NOTES:

1/ Formerly covered on NAVSHIPS Drawing 815-119711 and MS 17114. Symbol 234 has been superseded by Symbol 232. Symbol 230 (M16377/27-230) has been superseded by the L. C. Doane Co. Part Number 430, or equal.

2/ Former berth lights without receptacle (Symbol 229, NSN 6230-00-578-7547; Symbol 231, NSN 6230-00-808-4466; and Symbol 232, NSN 6230-00-578-7548), procured for replacement purposes for ships where isolated receptacle circuits have not been provided for berth lights, shall be in accordance with Symbols 229.1, 231.1, and 232.1, respectively as specified herein except that the receptacle shall not be furnished and the cable shall be a three-conductor, LSTHOF-3, in accordance with MIL-DTL-24643/3.

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – SH

Review activity:
DLA – IS

Preparing activity:
Navy – SH
(Project 6230-2012-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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